

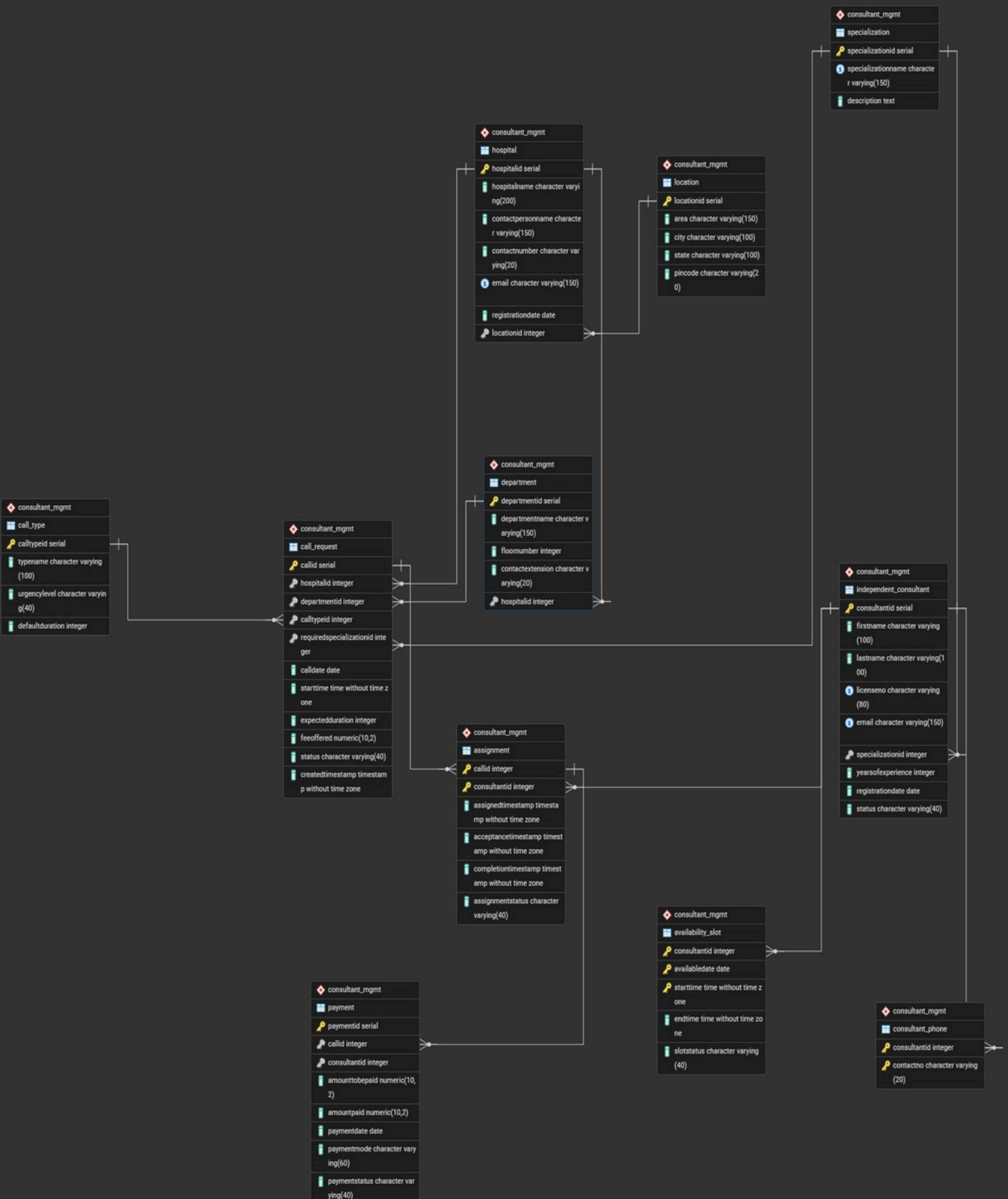


IT216 DATABASE MANAGEMENT SYSTEM

Independent Medical Consultant Coordination System(IMCCS)

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1. Normalization Relational schema Diagram



2. Minimal Functional Dependency Set

#	Relation	Functional Dependency
FD1	LOCATION	LocationID $\rightarrow$ Area
FD2	LOCATION	LocationID $\rightarrow$ City
FD3	LOCATION	LocationID $\rightarrow$ State
FD4	LOCATION	LocationID $\rightarrow$ PinCode
FD5	HOSPITAL	HospitalID $\rightarrow$ HospitalName
FD6	HOSPITAL	HospitalID $\rightarrow$ ContactPersonName
FD7	HOSPITAL	HospitalID $\rightarrow$ ContactNumber
FD8	HOSPITAL	HospitalID $\rightarrow$ Email
FD9	HOSPITAL	HospitalID $\rightarrow$ RegistrationDate
FD10	HOSPITAL	HospitalID $\rightarrow$ LocationID
FD11	HOSPITAL	Email $\rightarrow$ HospitalID
FD12	DEPARTMENT	DepartmentID $\rightarrow$ DepartmentName
FD13	DEPARTMENT	DepartmentID $\rightarrow$ FloorNumber

FD14	DEPARTMENT	DepartmentID → ContactExtension
FD15	DEPARTMENT	DepartmentID → HospitalID
FD16	SPECIALIZATION	SpecializationID → SpecializationName
FD17	SPECIALIZATION	SpecializationID → Description
FD18	SPECIALIZATION	SpecializationName → SpecializationID
FD19	INDEPENDENT_CONSULTANT	ConsultantID → FirstName
FD20	INDEPENDENT_CONSULTANT	ConsultantID → LastName
FD21	INDEPENDENT_CONSULTANT	ConsultantID → LicenseNo
FD22	INDEPENDENT_CONSULTANT	ConsultantID → Email
FD23	INDEPENDENT_CONSULTANT	ConsultantID → YearsOfExperience
FD24	INDEPENDENT_CONSULTANT	ConsultantID → RegistrationDate
FD25	INDEPENDENT_CONSULTANT	ConsultantID → Status
FD26	INDEPENDENT_CONSULTANT	ConsultantID → SpecializationID

FD27	INDEPENDENT_CONSULTANT	LicenseNo → ConsultantID
FD28	INDEPENDENT_CONSULTANT	Email → ConsultantID
FD29	CONSULTANT_PHONE	(ConsultantID, ContactNo) → (none)
FD30	CALL_TYPE	CallTypeID → TypeName
FD31	CALL_TYPE	CallTypeID → UrgencyLevel
FD32	CALL_TYPE	CallTypeID → DefaultDuration
FD33	AVAILABILITY_SLOT	(ConsultantID, AvailableDate, StartTime) → EndTime
FD34	AVAILABILITY_SLOT	(ConsultantID, AvailableDate, StartTime) → SlotStatus
FD35	CALL_REQUEST	CallID → HospitalID
FD36	CALL_REQUEST	CallID → DepartmentID
FD37	CALL_REQUEST	CallID → CallTypeID
FD38	CALL_REQUEST	CallID → RequiredSpecializationID
FD39	CALL_REQUEST	CallID → CallDate
FD40	CALL_REQUEST	CallID → StartTime
FD41	CALL_REQUEST	CallID → ExpectedDuration

FD42	CALL_REQUEST	CallID → FeeOffered
FD43	CALL_REQUEST	CallID → Status
FD44	CALL_REQUEST	CallID → CreatedTimestamp
FD45	ASSIGNMENT	(CallID, ConsultantID) → AssignedTimestamp
FD46	ASSIGNMENT	(CallID, ConsultantID) → AcceptanceTimestamp
FD47	ASSIGNMENT	(CallID, ConsultantID) → CompletionTimestamp
FD48	ASSIGNMENT	(CallID, ConsultantID) → AssignmentStatus
FD49	PAYMENT	PaymentID → CallID
FD50	PAYMENT	PaymentID → ConsultantID
FD51	PAYMENT	PaymentID → AmountToBePaid
FD52	PAYMENT	PaymentID → AmountPaid
FD53	PAYMENT	PaymentID → PaymentDate
FD54	PAYMENT	PaymentID → PaymentMode
FD55	PAYMENT	PaymentID → PaymentStatus

3. Proof that Relations are in BCNF

Relation	Candidate Key(s)	FDs Verified	Verdict
LOCATION	{LocationID}	FD1–FD4: LocationID→* ✓	IN BCNF
HOSPITAL	{HospitalID}, {Email}	FD5–FD11: HospitalID→* ✓; Email→HospitalID: Email is CK ✓	IN BCNF
DEPARTMENT	{DepartmentID}	FD12–FD15: DepartmentID→* ✓	IN BCNF
SPECIALIZATION	{SpecializationID}, {SpecializationName}	FD16–FD18: both CKs verified ✓	IN BCNF
INDEPENDENT_CONSULTANT	{ConsultantID}, {LicenseNo}, {Email}	FD19–FD28: ConsultantID→* ✓ ; LicenseNo→ConsultantID: LicenseNo is CK ✓; Email→ConsultantID: Email is CK ✓	IN BCNF
CONSULTANT_PHONE	{ConsultantID, ContactNo}	FD29: No non-trivial FDs; PK = entire relation ✓	IN BCNF
CALL_TYPE	{CallTypeID}	FD30–FD32: CallTypeID→* ✓	IN BCNF
AVAILABILITY_SLOT	{ConsultantID, AvailableDate, StartTime}	FD33–FD34: composite PK→* ✓; no partial/transitive deps ✓	IN BCNF

CALL_REQUEST	{CallID}	FD35–FD44: CallID→* ✓	IN BCNF
ASSIGNMENT	{CallID, ConsultantID}	FD45–FD48: (CallID,ConsultantID)→* ✓; no partial/transitive deps ✓	IN BCNF
PAYMENT	{PaymentID}	FD49–FD55: PaymentID→* ✓	IN BCNF

3.1 Detailed Proof — AVAILABILITY_SLOT (composite key, weak entity)

Relation: AVAILABILITY_SLOT (ConsultantID, AvailableDate, StartTime, EndTime, SlotStatus)

Step 1 — Candidate key computation:

- Closure({ConsultantID, AvailableDate, StartTime})+ :
 - Apply FD31–FD32: {ConsultantID, AvailableDate, StartTime} → EndTime, SlotStatus
 - Result: covers all 5 attributes ✓
 - No partial subset is a superkey:
 - Closure({ConsultantID})+ = {ConsultantID} only — does not determine slot attributes
 - Closure({ConsultantID, AvailableDate})+ = {ConsultantID, AvailableDate} only — a consultant could have multiple slots starting at different times on the same date
 - Closure({ConsultantID, StartTime})+ = {ConsultantID, StartTime} only — same start time on different dates is possible
- Therefore {ConsultantID, AvailableDate, StartTime} is the sole candidate key.

Step 2 — BCNF check for every FD:

FD	LHS Closure	LHS is Superkey?	BCNF?
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(ConsultantID, AvailableDate, StartTime) → EndTime	Covers all attributes	YES — is CK	✓ BCNF OK
(ConsultantID, AvailableDate, StartTime) → SlotStatus	Covers all attributes	YES — is CK	✓ BCNF OK

Conclusion: AVAILABILITY_SLOT is in BCNF. No decomposition required.

3.1 Detailed Proof — ASSIGNMENT (composite key, M:N resolution)

Relation: ASSIGNMENT (CallID, ConsultantID, AssignedTimestamp, AcceptanceTimestamp, CompletionTimestamp, AssignmentStatus)

Step 1 — Candidate key computation:

- Closure({CallID, ConsultantID})+ :
 - Apply FD43–FD46: {CallID, ConsultantID} → AssignedTimestamp, AcceptanceTimestamp, CompletionTimestamp, AssignmentStatus
 - Covers all 6 attributes ✓
 - No partial subset is a superkey: Closure({CallID})+ = {CallID} only; Closure({ConsultantID})+ = {ConsultantID} only. Neither alone covers all attributes.
- Sole candidate key: {CallID, ConsultantID}

Step 2 — BCNF check for every FD:

FD	LHS Closure	LHS is Superkey?	BCNF?
(CallID, ConsultantID) → AssignedTimestamp	Covers all attributes	YES — is CK	✓ BCNF OK
(CallID, ConsultantID) → AcceptanceTimestamp	Covers all attributes	YES — is CK	✓ BCNF OK
(CallID, ConsultantID) → CompletionTimestamp	Covers all attributes	YES — is CK	✓ BCNF OK

(CallID, ConsultantID) → AssignmentStatus	Covers all attributes	YES — is CK	✓ BCNF OK
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Conclusion: ASSIGNMENT is in BCNF. No decomposition required.

3.3 Detailed Proof — INDEPENDENT_CONSULTANT (multiple candidate keys)

Relation: INDEPENDENT_CONSULTANT (ConsultantID, FirstName, LastName, LicenseNo, Email, SpecializationID, YearsOfExperience, RegistrationDate, Status)

Step 1 — Candidate key computation:

- Closure({ConsultantID})+ applying FD19–FD26: covers all 9 attributes → {ConsultantID} is a candidate key ✓
- Closure({LicenseNo})+ : apply FD27 → get ConsultantID; then apply FD19–FD26 → covers all 9 attributes → {LicenseNo} is an alternate candidate key ✓
- Closure({Email})+ : apply FD28 → get ConsultantID; then apply FD19–FD26 → covers all 9 attributes → {Email} is an alternate candidate key ✓

Step 2 — BCNF check for every non-trivial FD:

FD	LHS Closure	LHS is Superkey?	BCNF?
ConsultantID → FirstName	Covers all attributes	YES — primary CK	✓ BCNF OK
ConsultantID → LastName	Covers all attributes	YES — primary CK	✓ BCNF OK
ConsultantID → LicenseNo	Covers all attributes	YES — primary CK	✓ BCNF OK
ConsultantID → Email	Covers all attributes	YES — primary CK	✓ BCNF OK
ConsultantID → SpecializationID	Covers all attributes	YES — primary CK	✓ BCNF OK

LicenseNo $\rightarrow$ ConsultantID	Covers all attributes	YES — alternate CK	✓ BCNF OK
Email $\rightarrow$ ConsultantID	Covers all attributes	YES — alternate CK	✓ BCNF OK

Conclusion: INDEPENDENT_CONSULTANT is in BCNF. The non-trivial FDs LicenseNo $\rightarrow$ ConsultantID and Email $\rightarrow$ ConsultantID do not violate BCNF because both LicenseNo and Email are themselves candidate keys.

3.4 Detailed Proof — CALL_REQUEST

Relation: CALL_REQUEST (CallID, HospitalID, DepartmentID, CallTypeID, RequiredSpecializationID, CallDate, StartTime, ExpectedDuration, FeeOffered, Status, CreatedTimestamp)

Step 1 — Candidate key:

- Closure($\{CallID\}$)⁺ applying FD35–FD44 covers all 11 attributes. $\{CallID\}$ is the sole candidate key.

Step 2 — BCNF check:

- All FDs FD35–FD44 have LHS = $\{CallID\}$. Since $\{CallID\}$ is the sole CK it is a superkey for every FD. All FDs satisfy BCNF.

Conclusion: CALL_REQUEST is in BCNF. No decomposition required.